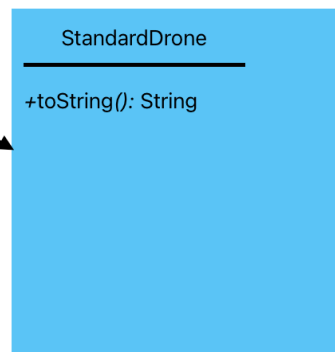
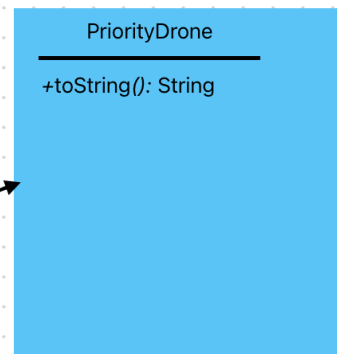
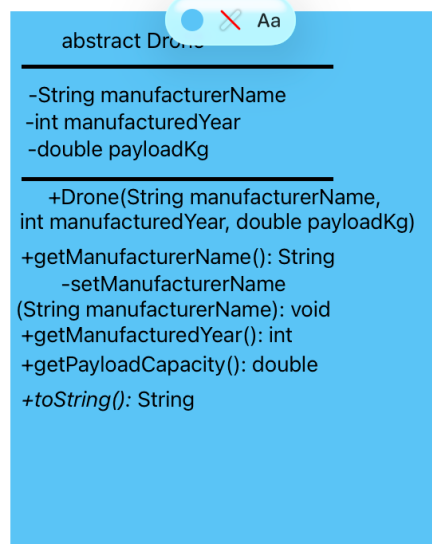
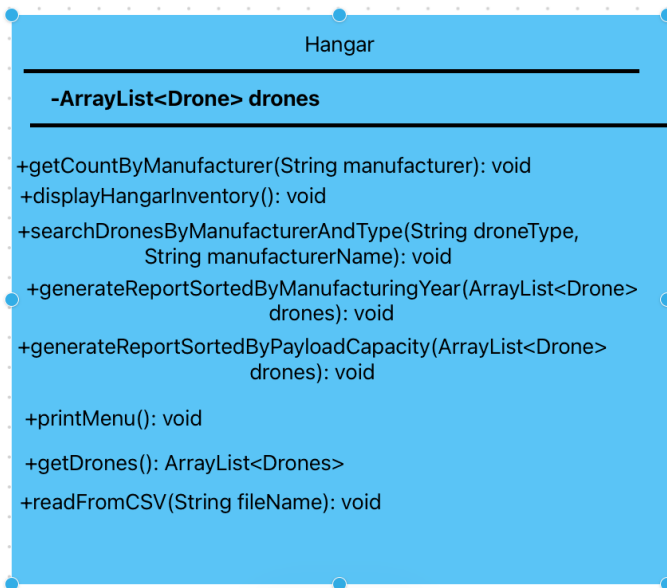




Purpose	Input / Setup	Expected Result
Verify valid drone records are loaded from a correctly formatted CSV file	CSV file contains: S,DJI,2021,12.5 and P,Skydio,2024,18.0	Two drone objects are added to the hangar inventory
Verify a CSV line with too few fields is skipped	CSV line: P,2024,18.0	Line is skipped and an error message is displayed
Verify a CSV line with an invalid drone type is skipped	CSV line: X,Parrot,2019,5.2	Line is skipped and an error message is displayed
Verify a CSV line with a non-numeric year or payload is skipped	CSV line: S,DJI,2021,10	Line is skipped and an error message is displayed
Verify a CSV line with a negative payload is skipped	CSV line: S,DJI,2021,-4.0	Line is skipped and an error message is displayed
Verify a CSV line with a missing manufacturer is skipped	CSV line: S,,2021,12.5	Line is skipped and an error message is displayed
Verify file-not-found handling works correctly	Enter a file name that does not exist	No drones are loaded and a file-not-found message is displayed
Verify counting works when a manufacturer has multiple drones	Inventory contains 2 DJI drones and 1 Autel drone; input manufacturer: DJI	Program reports a total count of 2
Verify counting works when a manufacturer has one drone	Inventory contains 1 Skydio drone; input manufacturer: Skydio	Program reports a total count of 1
Verify counting returns zero when the manufacturer is not present	Inventory contains DJI and Autel drones; input manufacturer: Parrot	Program reports a total count of 0
Verify manufacturer counting is case-insensitive	Inventory contains manufacturer DJI; input manufacturer: dji	Program reports the correct count for DJI
Verify payload sorting orders drones from lowest to highest	Inventory payloads: 12.5, 18.0, 5.2, 9.0	Sorted report displays payload order: 5.2, 9.0, 12.5, 18.0
Verify payload sorting does not change the original inventory order	Inventory is loaded in one order, then payload sort is run	Sorted report is correct, but original hangar inventory remains unchanged
Verify payload sorting handles an empty inventory without crashing	Inventory is empty	Sort report runs without errors and displays no drone records

Verify search finds matching priority drones for a manufacturer	Inventory contains a Priority Drone from Autel; input manufacturer: Autel, type: Priority	Matching priority drone is displayed
Verify search finds matching standard drones for a manufacturer	Inventory contains a Standard Drone from DJI; input manufacturer: DJI, type: Standard	Matching standard drone is displayed
Verify search supports abbreviated type input	Inventory contains a Priority Drone from Skydio; input manufacturer: Skydio, type: P	Matching priority drone is displayed
Verify search by manufacturer is case-insensitive	Inventory contains manufacturer Autel; input manufacturer: autel, type: Priority	Matching drone is displayed
Verify search reports no matches when none exist	Inventory contains only DJI drones; input manufacturer: Autel, type: Priority	Program displays a no-results message
Verify search rejects an invalid drone type	Input manufacturer: DJI, type: Express	Program displays an invalid drone type message